Shrabya Bhattarai

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EDUCATION

University of Southern Mississippi Hattiesburg, MS
Bachelor of Science in Computer Engineering *Aug 2024 – May 2028*

- GPA: 3.91/4.0 | Relevant Coursework: Digital Electronics, Computer Networks, Operating Systems, Embedded Systems, Data Structures

TECHNICAL SKILLS

Languages: Python, C, C++, SQL
Data & Analytics: Python scripting, SQL, data cleaning, CRM databases, automated reporting
AI & Vision: YOLO, vision-language models, model fine-tuning
Hardware: Arduino, sensors, motor drivers, Mecanum drive
Tools & Frameworks: Flask, Git

EXPERIENCE

4th Dimension Marketing Solutions Cincinnati, OH
Data Analyst Intern *May 2026 – Aug 2026*

- Write Python scripts that automate the collection and organization of daily field sales data into structured databases, replacing manual entry across multiple sales locations.
- Clean and validate raw sales records from physical marketing campaigns, resolving inconsistencies before the data enters reporting and analysis workflows.
- Build automated reporting pipelines that generate weekly sales performance summaries for the marketing team, reducing turnaround on reporting requests.
- Analyze multi-location sales data to identify performance trends and surface insights used to adjust campaign strategy.
- Maintain a centralized CRM database using SQL to track leads, sales figures, and campaign outcomes across the team.

TechMatrix Kathmandu, Nepal
Junior Software Engineer Intern *May 2023 – Aug 2024*

- Worked on the IoT side of the team's software projects, writing parts of the backend in Python and Flask.
- Collaborated with a small team of engineers on shared projects.

PROJECTS

Wildfire Smoke Detection on Edge Hardware Python, YOLO, Vision-Language Models
Research Project | University of Southern Mississippi *Jan 2026 – Present*

- Co-developing a two-stage system: a lightweight image classifier screens every camera frame, and a small vision-language model reviews only the frames it is unsure about.
- Put the paper together, found and reviewed the reference literature, and helped train the models.
- Currently helping debug the models; paper in progress.

Lost: Live Lecture Confusion App Node.js, SQLite, Claude API, Deepgram
VOXO Hatchathon 2026 | University of Southern Mississippi *Nov 2026*

- Built a web app where students tap the exact line of a live lecture transcript that lost them, and the professor sees those lines light up in real time.
- Used Deepgram speech-to-text for the live transcript and Claude for short explanations and after-class summaries drawn from the course materials.
- Added student and professor logins, a SQLite database and 28 automated tests, and pitched the app live to the judges.

Robo Soccer — Autonomous Arduino Robot Arduino, C++
Finalist | Google Developer Group (GDG) USM Competition *Oct 2025*

- Designed and built a custom Arduino-powered robot for autonomous soccer play.
- Programmed movement algorithms for ball tracking and precision control.
- Reached the finals among multiple university teams.

ACTIVITIES & LEADERSHIP

USM Open Source Community (OSC) University of Southern Mississippi
Social Media Lead & Co-Founder *March 2026 – Present*

- Co-founded the OSC club and manage the @oscusm social media presence, growing the community to 200+ student members through workshops, open-source contribution drives, and event coordination.